📘 **AI/ML Internship – Day 3 Notes & Tasks**

🟢 **Module 3: Python for AI**

📌 **Part 1: Python Collections**

🔹 **What are Collections?**

Collections are used to store multiple values in a single variable.

🔹 **Types of Collections**

- List
- Tuple
- Set
- Dictionary

📌 **Part 2: Lists**

🔹 **What is a List?**

A list is an ordered, changeable collection.

```
numbers = [1, 2, 3, 4, 5]
print(numbers)
```

🔹 **List Operations**

```
numbers.append(6)     # Add element
numbers.remove(2)     # Remove element
print(numbers[0])     # Access element
```

🔹 **Looping through List**

```
for num in numbers:
    print(num)
```

📌 Part 3: Tuples

🔹 What is a Tuple?

Tuple is similar to list but **cannot be changed**.

```
t = (1, 2, 3)
print(t)
```

📌 Part 4: Sets

🔹 What is a Set?

- Unordered collection
- No duplicate values

```
s = {1, 2, 3, 3}
print(s)
```

📌 Part 5: Dictionaries (Very Important for AI)

🔹 What is a Dictionary?

Stores data in key-value pairs.

```
student = {
   "name": "John",
   "age": 21,
   "marks": 90
}

print(student["name"])
```

🔹 Dictionary Operations

```
student["age"] = 22
student["grade"] = "A"
```

📌 Part 6: Strings

🔹 String Basics

```
text = "Hello AI"
print(text.lower())
print(text.upper())
```

🔹 String Operations

```
print(len(text))
print(text[0])
print(text[0:5])
```

📌 Part 7: List Comprehension (Important Concept)

```
numbers = [1, 2, 3, 4]

squares = [x*x for x in numbers]
print(squares)
```

🎯 Tasks for Day 3

🧠 Task 1: Theory

1. What is a list?
2. Difference between list and tuple
3. What is a dictionary?
4. What is a set?
5. What is list comprehension?

💻 Task 2: List Practice

Write programs:

1. Create a list of 10 numbers
2. Find sum of list elements
3. Find maximum number
4. Print even numbers from list

5. Reverse a list

💻 **Task 3: Dictionary Practice (Important)**

1. Create student dictionary (name, age, marks)
2. Update marks
3. Add new key (grade)
4. Print all keys and values

💻 **Task 4: String Practice**

1. Count number of characters
2. Reverse a string
3. Check palindrome
4. Count vowels

💻 **Task 5: Set Practice**

1. Remove duplicates from list using set
2. Find common elements between two sets

💻 **Task 6: List Comprehension (Important)**

1. Create list of squares from 1–10
2. Create list of even numbers from 1–20

🌐 **Task 7: Git Practice**

1. Create file:

day3.py

2. Add all programs
3. Push to GitHub:

```
git add .
git commit -m "Day 3 collections completed"
git push
```